Sturm–Liouville Problem

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What Is a Sturm–Liouville Problem?

- A Sturm–Liouville (S–L) problem (SLP) is a boundary value problem of the form

$$\frac{d}{dx}\left[p(x)\frac{dy}{dx}\right] + \left[q(x) + \lambda w(x)\right]y = 0, \quad a \le x \le b$$

together with a pair of linear homogeneous boundary conditions at $x = a$ and $x = b$. [Named after **Jacques Charles François Sturm** and **Joseph Liouville** (1830s).]
- Here $p(x), p'(x), q(x), w(x)$ are real and continuous on $[a, b]$.
- $p(x) > 0$ and $w(x) > 0$ on (a, b) (regular case).
- λ is a parameter, not given in advance.
- The problem: find the values of λ (**eigenvalues**) for which a nontrivial solution (**eigenfunction**) exists.
- $w(x)$ is called the **weight function**.



The General Second-Order Linear ODE

- Start from a general linear second order ODE with a parameter λ:

$$a_0(x)y'' + a_1(x)y' + \big[a_2(x) + \lambda a_3(x)\big]y = 0$$

- Here $a_0(x) \neq 0$ on (a, b).
- Such an equation is not, in general, self-adjoint.
- Self-adjointness is desirable because it guarantees real eigenvalues and orthogonal eigenfunctions.
- Every such equation can be converted into self-adjoint (Sturm–Liouville) form.
- This is done by multiplying through by a suitable integrating factor $p(x)$.
- The resulting form is $\dfrac{d}{dx}[py'] + (q + \lambda w)y = 0$.
- This reduction is the first key technical step in S–L theory.



Self-Adjoint Differential Operators

- A second order operator $L[y] = (py')' + qy$ is called **self-adjoint**.
- Self-adjointness means: for any two functions u, v satisfying the boundary conditions,

$$\int_a^b \left(u\,L[v] - v\,L[u]\right) dx = 0$$

- This is known as **Lagrange's identity**.
- In self-adjoint form, the boundary term reduces to

$$\Big[p(x)(uv' - vu')\Big]_a^b$$

- Self-adjointness is a structural property of p, q together with the boundary conditions.
- Not every ODE is self-adjoint in its original form; conversion is often required.
- Self-adjoint operators are the ODE analogue of symmetric



Standard Sturm–Liouville Form

- The standard (self-adjoint) Sturm–Liouville form is

$$\left[p(x)y'\right]' + \left[q(x) + \lambda w(x)\right]y = 0, \qquad a \le x \le b.$$

where $p(x) > 0$, $p'(x)$ continuous, $q(x)$ continuous, $w(x) > 0$ on $[a, b]$.

- Boundary conditions are linear and homogeneous, typically of the form

$$\alpha_1 y(a) + \alpha_2 y'(a) = 0, \qquad \beta_1 y(b) + \beta_2 y'(b) = 0,$$

with the conditions $(\alpha_1, \alpha_2) \neq (0, 0)$ and $(\beta_1, \beta_2) \neq (0, 0)$.

- Special cases: $\alpha_2 = 0$ gives a Dirichlet condition; $\alpha_1 = 0$ gives a Neumann condition.
- If $\alpha_1, \alpha_2, \beta_1, \beta_2$ are all nonzero, the condition is called mixed or Robin type.
- This is the canonical form used throughout our discussion.



Reduction to Self-Adjoint Form

- Given $a_0 y'' + a_1 y' + (a_2 + \lambda a_3) y = 0$, divide by $a_0(x)$:

$$y'' + \frac{a_1}{a_0} y' + \left(\frac{a_2}{a_0} + \lambda \frac{a_3}{a_0} \right) y = 0$$

- Multiply by the integrating factor

$$p(x) = \exp\left(\int \frac{a_1(x)}{a_0(x)} \, dx \right)$$

- The resulting equation is exactly of Sturm–Liouville form with

$$p(x) = p(x), \qquad q(x) = p(x) \frac{a_2(x)}{a_0(x)}, \qquad w(x) = p(x) \frac{a_3(x)}{a_0(x)}$$

- Verification: $(py')' = py'' + p'y' = p\left(y'' + \frac{a_1}{a_0} y' \right)$, matching the multiplied equation.
- This construction always works locally, wherever $a_0(x) \neq 0$.



Example: Reducing a General Equation

- Consider $xy'' + y' + \lambda xy = 0$ on $x > 0$ (this is the parametric Bessel equation, order $n = 0$).
- Here $a_0 = x,\ a_1 = 1,\ a_2 = 0,\ a_3 = x$.
- Integrating factor: $p(x) = \exp\left(\int \frac{1}{x} dx\right) = x$.
- Multiplying by $p/a_0 = x/x = 1$ leaves the equation unchanged: $xy'' + y' + \lambda xy = 0$.
- Rewrite as $(xy')' + \lambda xy = 0$.
- So $p(x) = x,\ q(x) = 0,\ w(x) = x$ — already self-adjoint!
- This equation is singular at $x = 0$ since $p(0) = 0$: a **singular** S–L problem.



Regular Sturm–Liouville Problem

- A S–L problem on $[a, b]$ is called **regular** if:
- (i) $p(x) > 0$ and $w(x) > 0$ for all $x \in [a, b]$ (including endpoints);
- (ii) p, p', q, w are continuous on $[a, b]$;
- (iii) the interval $[a, b]$ is finite;
- (iv) the boundary conditions are separated (one condition at a, one at b).
- Regular problems have the cleanest theory: real, simple, countably infinite eigenvalues.
- Example: $y'' + \lambda y = 0$, $y(0) = y(L) = 0$ is regular.
- Most classical trigonometric eigenfunction problems are regular.



Singular Sturm–Liouville Problem

- A problem is **singular** if any regularity condition fails, i.e.,
- For example: (i) $p(x) \to 0$ at an endpoint (as with Legendre's or the parametric Bessel equation);
- (ii) the interval is infinite or semi-infinite (as with Hermite's or Laguerre's equation);
- (iii) $q(x)$ or $w(x)$ is unbounded at an endpoint.
- At a singular endpoint, instead of a boundary condition we typically require boundedness (finiteness) of y.
- Example: parametric Bessel equation on $[0, a]$ — singular at $x = 0$; require y finite there.
- Example: Legendre's equation on $[-1, 1]$ — singular at $x = \pm 1$; require y finite there.
- Example: Hermite's equation on $(-\infty, \infty)$ — singular due to the infinite interval.
- Despite added technical difficulty, singular S–L problems still enjoy orthogonality of eigenfunctions.



Periodic Sturm–Liouville Problem

- A third type: **periodic** boundary conditions, e.g. on $[-L, L]$:

$$y(-L) = y(L), \qquad y'(-L) = y'(L)$$

- Also requires $p(-L) = p(L)$ for self-adjointness to hold.
- Example: $y'' + \lambda y = 0$ with periodic BCs gives the classical Fourier series (sines *and* cosines).
- Unlike the regular case, periodic problems can have eigenvalues of multiplicity two.
- Eigenfunctions are still orthogonal with respect to the weight $w(x)$.
- This case underlies full (not half-range) Fourier series expansions.



Types of Boundary Conditions

- **Dirichlet:** $y(a) = 0,\ y(b) = 0.$
- **Neumann:** $y'(a) = 0,\ y'(b) = 0.$
- **Mixed (Robin):** $\alpha y(a) + \alpha' y'(a) = 0$, similarly at b.
- **Periodic:** $y(a) = y(b),\ y'(a) = y'(b).\ [p(a) = p(b)]$
- **Singular / natural:** y (and py') required to remain finite at a singular endpoint.
- Each type leads to different transcendental or algebraic equations for the eigenvalues.
- The nature of the boundary conditions determines completeness of the eigenfunction set.



The Eigenvalue Problem

- Given the S–L equation $(py')' + (q + \lambda w)y = 0$ with boundary conditions $B_1[y] = 0,\ B_2[y] = 0$.
- A value λ for which a *nontrivial* solution $y(x)$ exists is called an **eigenvalue**.
- The corresponding nontrivial solution $y(x)$ is called an **eigenfunction**.
- The trivial solution $y \equiv 0$ always satisfies the problem, but is excluded by definition.
- Eigenfunctions are determined only up to a nonzero scalar multiple.
- The set of all eigenvalues is called the **spectrum** of the problem.
- Finding eigenvalues generally reduces to solving a transcendental equation from the boundary conditions.



Trivial Solutions and Eigenvalues

- For most λ, the only solution satisfying both boundary conditions is $y \equiv 0$.
- Only for special, discrete values of λ does a nontrivial solution exist.
- This mirrors the matrix eigenvalue problem $Av = \lambda v$: nontrivial v exists only for special λ.
- The S–L operator $L[y] = -(py')' - qy$ acting on functions satisfying the BCs is the continuous analogue of a symmetric matrix.
- $L[y] = \lambda wy$ is the operator form of the S–L equation.
- This operator viewpoint explains why the theory parallels linear algebra so closely (real eigenvalues, orthogonal eigenvectors).



Existence Theorem (Simmons)

- **Theorem (Simmons, Ch. 11).** A regular S–L problem has an infinite sequence of real eigenvalues

$$\lambda_1 < \lambda_2 < \lambda_3 < \cdots \to \infty$$

- Each eigenvalue λ_n has (up to scalar multiples) a unique eigenfunction $y_n(x)$.
- The eigenfunction $y_n(x)$ has exactly $n-1$ zeros in the open interval (a, b) (oscillation theorem).
- There is no largest eigenvalue: $\lambda_n \to +\infty$ as $n \to \infty$.
- This parallels the existence of infinitely many normal modes of a vibrating string.
- The proof relies on comparing the S–L equation with $y'' + \lambda y = 0$ via Sturm comparison theorems.



Reality of Eigenvalues

- **Claim:** all eigenvalues of a (regular or singular, self-adjoint) S–L problem are real.
- **Proof sketch:** suppose $\lambda = p + i\nu$ is a complex eigenvalue with eigenfunction y.
- Then $\bar{\lambda}$ is an eigenvalue with eigenfunction $\bar{y}$ (conjugate the equation; coefficients are real).
- Using Lagrange's identity with $u = y,\ v = \bar{y}$, the boundary term vanishes by the boundary conditions.
- This forces $(\lambda - \bar{\lambda}) \int_a^b w|y|^2\, dx = 0$.
- Since $w > 0$ and $y \not\equiv 0$, the integral is strictly positive, so $\lambda = \bar{\lambda}$, i.e. λ is real.
- Hence all eigenvalues of a self-adjoint S–L problem must be real.



Case A: Dirichlet Boundary Conditions — Setup

- Classical S–L problem: $y'' + \lambda y = 0, \quad 0 \le x \le L.$
- Boundary conditions: $y(0) = 0, \quad y(L) = 0.$
- Here $p(x) = 1, \ q(x) = 0, \ w(x) = 1$: a regular S–L problem.
- We examine three cases for λ: negative, zero, and positive.
- This is the model problem behind the classical Fourier sine series.
- Physically: this describes the normal modes of a string fixed at both ends.



Case A: Solving for $\lambda \leq 0$

- Case (i) : If $\lambda = 0$: general solution $y = c_1 + c_2 x$. Applying $y(0) = 0$ gives $c_1 = 0$; $y(L) = 0$ gives $c_2 L = 0 \Rightarrow c_2 = 0$.
- So $y \equiv 0$: $\lambda = 0$ is not an eigenvalue.
- Case (ii): If $\lambda = -p^2 < 0$, AE is $m^2 - p^2 = 0$. Then $y(x) = c_1 e^{px} + c_2 e^{-px}$. (Use BCs to show that $c_1 = c_2 = 0$ So, $y \equiv 0$) - HOMEWORK.
- **OR**: General solution $y = c_1 \cosh(px) + c_2 \sinh(px)$.
- $y(0) = 0 \Rightarrow c_1 = 0$, so $y = c_2 \sinh(px)$.
- $y(L) = 0 \Rightarrow c_2 \sinh(pL) = 0$. Since $\sinh(pL) \neq 0$ for $p \neq 0$, $c_2 = 0$.
- So $y \equiv 0$: negative λ gives no eigenvalues.
- Conclusion: all eigenvalues must be strictly positive.



Case A: Solving for $\lambda > 0$ — Eigenvalues

- Case (iii): Let $\lambda = p^2 > 0$. General solution: $y = c_1 \cos(px) + c_2 \sin(px)$.
- $y(0) = 0 \Rightarrow c_1 = 0$, so $y = c_2 \sin(px)$.
- $y(L) = 0 \Rightarrow c_2 \sin(pL) = 0$.
- For a nontrivial solution, $c_2 \neq 0$, so we need $\sin(pL) = 0$.
- This gives $pL = n\pi$, i.e. $p_n = n\pi/L$, for $n = 1, 2, 3, \ldots$
- Hence the eigenvalues are

$$\lambda_n = \left(\frac{n\pi}{L}\right)^2, \qquad n = 1, 2, 3, \ldots$$



Case A: Eigenfunctions and Orthogonality

- The eigenfunctions are $y_n(x) = \sin\left(\dfrac{n\pi x}{L}\right)$, $n = 1, 2, 3, \ldots$
- Orthogonality (weight $w = 1$):

$$\int_0^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = 0, \quad m \neq n$$

- Norm: $\displaystyle\int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{L}{2}$.
- Normalized eigenfunctions: $\phi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$.
- This recovers the classical Fourier sine series as an eigenfunction expansion.
- This is the simplest and most important example of a regular S–L problem.



Case B: Neumann Boundary Conditions

- Same equation $y'' + \lambda y = 0$, now with $y'(0) = 0,\ y'(L) = 0$.
- As before, $\lambda \leq 0$ yields only the trivial solution except $\lambda = 0$: check separately.
- For $\lambda = 0$: $y = c_1 + c_2 x$, so $y' = c_2$. $y'(0) = 0 \Rightarrow c_2 = 0$; $y'(L) = 0$ automatically satisfied.
- So $y = c_1$ (constant) is a valid eigenfunction for $\lambda_0 = 0$.
- For $\lambda = p^2 > 0$: $y = c_1 \cos(px) + c_2 \sin(px)$, $y' = -c_1 p \sin(px) + c_2 p \cos(px)$.
- $y'(0) = 0 \Rightarrow c_2 = 0$, so $y = c_1 \cos(px)$.
- $y'(L) = 0 \Rightarrow -c_1 p \sin(pL) = 0 \Rightarrow \sin(pL) = 0 \Rightarrow p_n = n\pi/L$.



Case B: Eigenvalues and Eigenfunctions

- Eigenvalues: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $n = 0, 1, 2, 3, \ldots$ (including $n = 0$).
- Eigenfunctions: $y_n(x) = \cos\left(\frac{n\pi x}{L}\right)$, $n = 0, 1, 2, \ldots$
- Orthogonality: $\int_0^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = 0$ for $m \neq n$.
- Norms: $\|y_0\|^2 = L$; $\|y_n\|^2 = L/2$ for $n \geq 1$.
- This gives rise to the classical Fourier cosine series.
- Physically: this models a string (or rod) with free ends.



Case C: Mixed BC — Dirichlet–Neumann

- $y'' + \lambda y = 0$, with $y(0) = 0, \quad y'(L) = 0.$
- $\lambda \leq 0$ again gives only the trivial solution (similar argument as Case A).
- For $\lambda = p^2 > 0$: $y = c_1 \cos(px) + c_2 \sin(px)$.
- $y(0) = 0 \Rightarrow c_1 = 0$, so $y = c_2 \sin(px)$, $y' = c_2 p \cos(px)$.
- $y'(L) = 0 \Rightarrow c_2 p \cos(pL) = 0 \Rightarrow \cos(pL) = 0$.
- This gives $pL = \dfrac{(2n-1)\pi}{2}$, i.e. $p_n = \dfrac{(2n-1)\pi}{2L}$, $n = 1, 2, 3, \ldots$



Case C: Eigenvalues and Eigenfunctions

- Eigenvalues: $\lambda_n = \left(\dfrac{(2n-1)\pi}{2L}\right)^2$, $n = 1, 2, 3, \ldots$
- Eigenfunctions: $y_n(x) = \sin\left(\dfrac{(2n-1)\pi x}{2L}\right)$.
- These form an orthogonal system on $[0, L]$ with weight $w = 1$.
- Norm: $\displaystyle\int_0^L y_n(x)^2\,dx = L/2$ for every n.
- Physically models a string fixed at $x = 0$ and free at $x = L$.
- Only odd "quarter-wave" harmonics appear.



Case D: Mixed BC — Neumann–Dirichlet

- $y'' + \lambda y = 0$, with $y'(0) = 0, \quad y(L) = 0.$
- For $\lambda = p^2 > 0$: $y = c_1 \cos(px) + c_2 \sin(px)$,
  $y' = -c_1 p \sin(px) + c_2 p \cos(px)$.
- $y'(0) = 0 \Rightarrow c_2 = 0$, so $y = c_1 \cos(px)$.
- $y(L) = 0 \Rightarrow c_1 \cos(pL) = 0 \Rightarrow \cos(pL) = 0$.
- This again gives $p_n = \dfrac{(2n-1)\pi}{2L}$, $n = 1, 2, 3, \ldots$
- Negative and zero λ again give only the trivial solution.



Case D: Eigenvalues and Eigenfunctions

- Eigenvalues: $\lambda_n = \left(\dfrac{(2n-1)\pi}{2L}\right)^2$, $n = 1, 2, 3, \ldots$ (same spectrum as Case C).
- Eigenfunctions: $y_n(x) = \cos\left(\dfrac{(2n-1)\pi x}{2L}\right)$.
- Orthogonal on $[0, L]$ with weight $w = 1$; norm $L/2$.
- Physically models a rod free at $x = 0$ and fixed at $x = L$.
- Cases C and D are "mirror images" of one another.
- Together Cases A–D illustrate how each boundary-condition type reshapes the eigenvalue spectrum.



Case E: Periodic Boundary Conditions

- $y'' + \lambda y = 0$ on $[-L, L]$, with $y(-L) = y(L)$ and $y'(-L) = y'(L)$.
- As before, $\lambda \leq 0$ gives only trivial or constant solutions; check $\lambda = 0$ separately.
- $\lambda = 0$: $y = c_1 + c_2 x$. Periodicity forces $c_2 = 0$; $y = c_1$ is an eigenfunction with $\lambda_0 = 0$.
- For $\lambda = p^2 > 0$: $y = c_1 \cos(px) + c_2 \sin(px)$.
- Imposing both periodicity conditions leads to $\sin(pL) = 0$, i.e. $p_n = n\pi/L$.
- Unlike Cases A–D, here *both* $\cos(p_n x)$ and $\sin(p_n x)$ are eigenfunctions for the same λ_n ($n \geq 1$): multiplicity 2.



Case E: Eigenvalues and Eigenfunctions

- Eigenvalues: $\lambda_n = \left(\frac{n\pi}{L}\right)^2$, $n = 0, 1, 2, \ldots$
- For $n = 0$: eigenfunction $y_0 = 1$.
- For $n \geq 1$: two independent eigenfunctions $\cos(n\pi x/L)$ and $\sin(n\pi x/L)$.
- All are mutually orthogonal on $[-L, L]$ with weight $w = 1$.
- This is exactly the eigenfunction system behind the full (two-sided) Fourier series.
- Periodic S–L problems are the natural setting for functions on a circle (angular variable θ).



Case F: Robin (Mixed) Boundary Conditions

- $y'' + \lambda y = 0$, $0 \le x \le L$, with $y(0) = 0$ and $y'(L) + h\,y(L) = 0$, $h > 0$ constant.
- Physically models a bar with one end fixed, the other in elastic (spring-like) support.
- For $\lambda = p^2 > 0$: $y(0) = 0 \Rightarrow y = c_2 \sin(px)$.
- Compute $y'(L) + hy(L) = c_2\left[p\cos(pL) + h\sin(pL)\right] = 0$.
- Nontrivial solution requires $p\cos(pL) + h\sin(pL) = 0$.



Case F: Transcendental Eigenvalue Equation

- The eigenvalue condition can be rewritten as

$$\tan(pL) = -\frac{p}{h}$$

- This is a **transcendental equation**: it cannot be solved in closed form for p.
- It has infinitely many positive roots $p_1 < p_2 < p_3 < \cdots$, found graphically or numerically.
- Eigenvalues are $\lambda_n = p_n^2$; eigenfunctions are $y_n(x) = \sin(p_n x)$.
- Despite the transcendental (non-explicit) eigenvalues, orthogonality of $\{y_n\}$ still holds (general theorem, Slide 22).
- This case is typical of "Robin" (mixed, elastic-support) physical boundary conditions.



The Cauchy–Euler Equation

- The Cauchy–Euler (equidimensional) equation has the form
$$x^2y'' + \alpha xy' + \beta y = 0$$
- Characterized by coefficients that are powers of x matching the order of differentiation.
- Solved by the trial substitution $y = x^m$, giving the **indicial equation** $m(m-1) + \alpha m + \beta = 0$.
- A **parametric Cauchy–Euler S–L equation** introduces λ:
$$x^2y'' + xy' + \lambda y = 0$$
- This equation is singular at $x = 0$ but regular on any interval $[a, b]$ with $0 < a < b$.
- It arises, e.g., from Laplace's equation in polar coordinates (radial part with $\partial^2/\partial\theta^2$ eigenvalue).



Cauchy–Euler Equation in Self-Adjoint Form

- Take $x^2y'' + xy' + \lambda y = 0$ on $[a, b] \subset (0, \infty)$. Divide by x:

$$xy'' + y' + \frac{\lambda}{x}y = 0 \quad \Longrightarrow \quad (xy')' + \frac{\lambda}{x}y = 0$$

- Comparing to $(py')' + (q + \lambda w)y = 0$: identify $p(x) = x,\ q(x) = 0,\ w(x) = 1/x$.
- Since $p(x) = x > 0$ on $[a, b]$ with $a > 0$, this is a **regular** S–L problem on $[a, b]$.
- The weight function here is $w(x) = 1/x$, not 1 — orthogonality will be with respect to this weight.
- This is the S–L structure underlying Cauchy–Euler eigenvalue problems.



Reduction via $x = e^t$

- The Cauchy–Euler equation is converted to constant-coefficient form by $x = e^t$, i.e. $t = \ln x$.
- Chain rule: $\dfrac{dy}{dx} = \dfrac{1}{x}\dfrac{dy}{dt}$, and $\dfrac{d^2y}{dx^2} = \dfrac{1}{x^2}\left(\dfrac{d^2y}{dt^2} - \dfrac{dy}{dt}\right)$.
- Substituting into $x^2y'' + xy' + \lambda y = 0$:

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} + \frac{dy}{dt} + \lambda y = 0 \implies \frac{d^2y}{dt^2} + \lambda y = 0$$

- This is exactly the constant-coefficient equation of Case A, now in the variable $t = \ln x$!
- So the Cauchy–Euler S–L problem is the trigonometric case in disguise, via a logarithmic change of variable.
- This substitution is the standard technique (Kreyszig, Ch. 2) for solving Cauchy–Euler equations.



Cauchy–Euler S–L Problem — Setup

- Consider $x^2y'' + xy' + \lambda y = 0$ on $[1, e]$, with boundary conditions $y(1) = 0,\ y(e) = 0$.
- Using $t = \ln x$: the interval $x \in [1, e]$ maps to $t \in [0, 1]$.
- Boundary conditions become $y(t = 0) = 0,\ y(t = 1) = 0$ — exactly Case A with $L = 1$.
- This is a natural "Cauchy–Euler analogue" of the classical trigonometric Dirichlet problem.
- We now solve directly in x, then verify consistency with the t-substitution.



Cauchy–Euler S–L Problem — Solving

- In terms of $t = \ln x$, we can write the solution as $y(t) = c_1 \cos(\sqrt{\lambda}\, t) + c_2 \sin(\sqrt{\lambda}\, t)$ for $\lambda > 0$.
- Changing to x: $y(x) = c_1 \cos(\sqrt{\lambda}\, \ln x) + c_2 \sin(\sqrt{\lambda}\, \ln x)$.
- As in Case A, $\lambda \leq 0$ yields only the trivial solution (same argument, translated via $t = \ln x$).
- Apply $y(1) = 0$: since $\ln 1 = 0$, this gives $c_1 = 0$.
- So $y(x) = c_2 \sin(\sqrt{\lambda}\, \ln x)$.
- Apply $y(e) = 0$: since $\ln e = 1$, this gives $\sin(\sqrt{\lambda}) = 0$, so $\sqrt{\lambda} = n\pi$.



Cauchy–Euler S–L Problem — Eigenvalues/Eigenfunctions

- Eigenvalues: $\lambda_n = (n\pi)^2$, $n = 1, 2, 3, \ldots$
- Eigenfunctions: $y_n(x) = \sin(n\pi \ln x)$, on $[1, e]$.
- These solve $x^2 y_n'' + x y_n' + (n\pi)^2 y_n = 0$ with $y_n(1) = y_n(e) = 0$.
- Compare with Case A eigenfunctions $\sin(n\pi t)$: identical, under $t = \ln x$.
- This illustrates a general principle: Cauchy–Euler S–L problems reduce to trigonometric S–L problems via $t = \ln x$.
- Any interval $[a, b]$ with $0 < a < b$ can be handled the same way, with $L = \ln(b/a)$.



Second Cauchy–Euler Example — Setup

- Consider $x^2y'' + xy' + \lambda y = 0$ on $[1, b]$, with $y(1) = 0$ and $y'(b) = 0$ (Robin-type at b).
- Using $t = \ln x \in [0, \ln b]$, let $M = \ln b$. In the t-variable, this is Case C with $L = M$.
- So the eigenvalues in t are $\sqrt{\lambda_n} = \dfrac{(2n-1)\pi}{2M}$.
- We must be careful: $y'(x) = \dfrac{1}{x}\dfrac{dy}{dt}$, so $y'(b) = 0 \iff \left.\dfrac{dy}{dt}\right|_{t=M} = 0$ — confirming the Neumann-type condition transfers directly.
- This illustrates how to handle a general Cauchy–Euler S–L problem with mixed boundary types.



Second Cauchy–Euler Example — Solution

- Eigenvalues: $\lambda_n = \left(\dfrac{(2n-1)\pi}{2\ln b}\right)^2$, $n = 1, 2, 3, \ldots$
- Eigenfunctions: $y_n(x) = \sin\left(\dfrac{(2n-1)\pi \ln x}{2\ln b}\right)$, on $[1, b]$.
- Orthogonal on $[1, b]$ with weight $w(x) = 1/x$.
- As $b \to e$, this reduces to a rescaled version of the previous example.
- This shows the Cauchy–Euler variety is entirely governed by the same catalogue of cases (A–F) as the trigonometric variety, via $t = \ln x$.

